Temperature and Humidity Monitoring System

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Aim:

To design and implement a temperature and humidity monitoring system using Arduino Uno and DHT11 sensor, with an optional integration of IoT functionality, to demonstrate real-time data acquisition and remote monitoring capabilities.

Requirements:

1. Arduino Uno board.
2. DHT11 Temperature and Humidity Sensor.
3. Breadboard and jumper wires.
4. USB cable for Arduino Uno.
5. Computer with Arduino IDE installed.
6. Internet connection (optional, for IoT functionality).

Procedure:

1. **Setup Hardware:**
   - Connect the DHT11 sensor to the Arduino Uno board using jumper wires.
   - Connect the VCC pin of the DHT11 to 5V on the Arduino, GND to GND, and the data pin to digital pin 2.
2. **Write Program:**
   - Open the Arduino IDE on your computer.
   - Write a program to read temperature and humidity data from the DHT11 sensor.

- Use the DHT library for Arduino to simplify the process.
- Add code to display the temperature and humidity values on the serial monitor.

3. **Test Program:**
   - Upload the program to the Arduino board.
   - Open the serial monitor in the Arduino IDE to view the temperature and humidity readings.
4. **Add IoT Functionality (Optional):**
   - If you want to make this a true IoT project, you can add connectivity to the internet.
   - Use an IoT platform like ThingSpeak or Blynk to send the sensor data to the cloud.
   - Modify your Arduino program to include code for connecting to Wi-Fi (if using) and sending data to the IoT platform.
5. **Monitor Data:**
   - Access the IoT platform dashboard to monitor the temperature and humidity data remotely (if applicable).

Program:

```
#include <DHT.h>


#define DHTPIN 2

#define DHTTYPE DHT11
```

```
DHT dht(DHTPIN, DHTTYPE);

void setup() {
Serial.begin(9600);
dht.begin();
}

void loop() {
delay(2000);

float temperature = dht.readTemperature();
float humidity = dht.readHumidity();

Serial.print("Temperature: ");
Serial.print(temperature);
Serial.print(" °C\t");
Serial.print("Humidity: ");
Serial.print(humidity);
Serial.println(" %");
}
```

Output:

Temperature: 25.00 °C Humidity: 50.00 %

Temperature: 25.10 °C Humidity: 51.00 %

Temperature: 25.20 °C Humidity: 52.00 %

... (continues)

Conclusions:

- The project successfully reads temperature and humidity data from the DHT11 sensor.
- The readings are displayed accurately on the serial monitor.
- If IoT functionality was added, the data can be monitored remotely through an IoT platform.
- This project serves as a basic foundation for more complex IoT projects involving sensor data monitoring and cloud connectivity.